BIOCHEMISTRY CUMULATIVE EXAM

April 5, 1997

Purification of biological molecules such as proteins is essential to study of their structure and function as many types of studies can only be accomplished with pure or homogeneous material. This examination focuses on some common concepts and techniques involved in protein purification.

(30 points) In designing a purification scheme for a protein, one will want to establish conditions under which the protein of interest is stable and can be purified. Discuss at least three factors which must be included in establishing such conditions and how they can be controlled.

(30 points) Several different types of chromatography are often used in protein purification, including ion exchange chromatography, gel permeation chromatography, hydrophobic chromatography, and affinity chromatography. For three of these types of chromatographic procedures, discuss their use in protein purification including the types of chromatographic materials used, the means by which they separate proteins, and loading and elution conditions.

(30 points) In purifying the protein of interest, you will generally be starting with a crude extract which contains hundreds to thousands of proteins. During your purification scheme you are going to have to follow the protein of interest. Discuss three ways in which this might be done.

(10 points) In the end, you will hopefully have a pure sample which contains only the protein of interest. Describe two ways in which you might demonstrate that this is indeed the case.